Hoosick Street and Burdett Avenue Redesign

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April 20th, 2025

CIVL - 2030 Introduction to Transportation Engineering

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1. OBJECTIVE

Car dependability in the United States (US) defines the country's main mode of transportation. Decades of urban planning in the country have been centered around automobiles. In the early 20th century, it became the main mode of transportation after World War II. The turning point for car dependency in the US was in the 1950s, the government made large investments in road infrastructure. When the Federal Aid Act was passed, it launched a newly reformed Highway System, cementing the use of cars as the main mode of transportation in the US. Consequently, car dependency in the United States has influenced social and environmental inequities.

Troy, NY reflects the industrial history of the city's evolution throughout the 19th century. The Troy and Schenectady Turnpike was built to enhance trade throughout the region. However, as industries began to leave Troy, the subsequent population decline had a detrimental effect on the city's infrastructure, including its streets. However, for Troy to become a more sustainable and equitable city, it needs to rethink its approach to transportation to have a city that is more pedestrian friendly and accommodating of people from all socio-economic backgrounds.

Congestion and speeding cars are some negative aspects of both the welfare of a community and the global environment. While these factors seem contradictory on the surface, when traffic is congested drivers are more susceptible to becoming frustrated, leading to speeding aggressively when they get the opportunity to move. The constant speeding and stopping can lead to rear-end collisions due to the lack of reaction and awareness. There are also speeding cars that weave through traffic, which causes accidents due to a reduction in reaction times and the need for increased stopping distance because of the higher speeds. The combination of unpredictable braking, lane switching, various speeds, and frustration turns it into a place of high risk for safety in that environment.

In current-day Troy, there are many congested roads where cars, trucks, buses, bikes, and pedestrians try to use roads designed with cars and trucks in mind. Consequently, the efficiency with which these roads allow people to traverse the city and the surrounding region is harmed. According to a report by the WAMC Northeast Public Radio, residents in Troy are frustrated with the congestion on Hoosick Street. In their report, many residents who live near Hoosick St try to avoid this street because of its notorious congestion and speeding issues. However, many residents still choose to use Hoosick because of its proximity to the many businesses that are near their homes.[1]

Hoosick Street is one of the busiest streets in Troy. According to the Capital District Transportation Committee (CDTC), The Average Annual Daily Traffic (AADT) of 10th to 15th street is 29,000 vehicles/day with 2,250 vehicles in the peak hour, around 19% of which are heavy vehicles, which can

lead to congestion, crashes and heavy pedestrian safety concerns. According to the same study done by CDTC, from March 1, 2014, to February 28, 2019, there were a total of 1,297 crashes that consisted of pedestrians, bicycles, and vehicles, leading to a pedestrian fatality on Hoosick. Even though this study spans 6 years there should not be that many collisions on any street.[2] The city of Troy is aware of these statistics, they so far have not developed a safer design of streets to mitigate these numbers.
Thus, the objective of this design project is to make Hoosick Street from 10th to Burdett Avenue more accessible and safe for Troy commuters, which will be achieved by improving and adding to the existing transportation infrastructure.

2. METHODOLOGY

The objective of this design project is to increase the safety of Hoosick Street from 10th Street - Burdett Ave. The main factors of this design are to mitigate pedestrian safety, encourage cycling and public transit use, and relieve congestion. Currently, Hoosick Street is a 4-lane street, two lanes in each direction, as shown in Fig. (1). There are two proposed changes for this street; one lane on each side will be converted to a bus-only lane, and a road verge will be added between the street and the sidewalk. On Burdett Avenue, a two-way bike lane will be installed as well.

The Capital District Transportation Authority (CDTA) services both Hoosick St. and Burdett Ave. with the Route 87 bus. Currently, at peak hours, the bus runs every 20 minutes.[3] Implementing a bus dedicated bus lane - as seen in Fig. (2) - should improve the reliability of the system, which should allow the intervals to decrease to 10 to 15 minutes. These changes are justified because the area Route 87 serves - the neighborhoods of Downtown Troy, Hillside, Frear Park, Sycaway, and the RPI campus - is home to a population with a high potential of shifting their transportation mode from car to bus.[4] These factors include individuals who do not have do not persons in care and live near a high-frequency public transit stop.[5] Supplementing this are the affordable price to ride the 87 Route, as each ride costs $1.50. A road verge was added to improve the visual aesthetics of Hoosick Street, which would attract a greater volume of pedestrians, while also ensuring that the sidewalks would be safer for altered street plans.

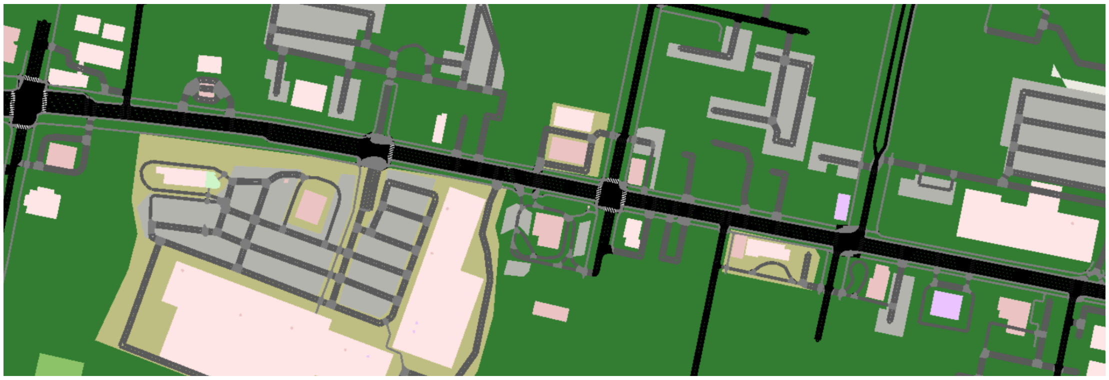

Figure 1. Hoosick Avenue before changes

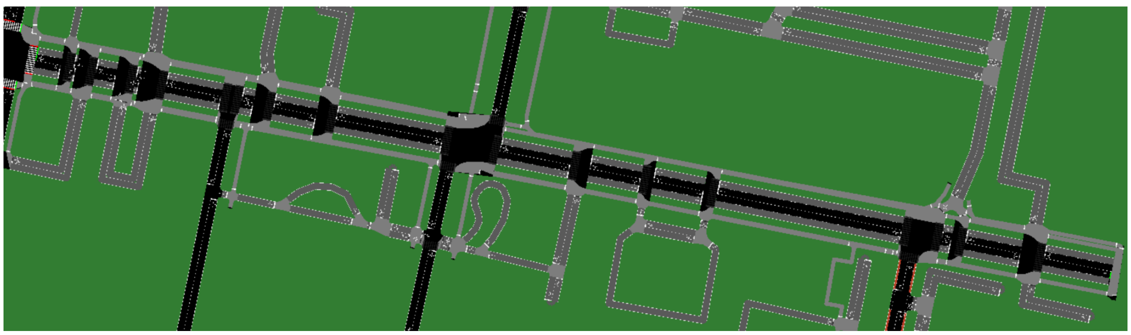

Figure 2. Proposed Hoosick St with Grey Bus Lanes

Burdett Ave., shown in Fig. (3), comparatively is less congested and has a lower heavy vehicle count than Hoosick St., so a bus lane does not need to be implemented. This opens up an opportunity for the addition of a two-way bike lane. This bike lane would serve the student population of RPI, who could better traverse between the East Campus Athletic Village and the main campus while opening up another mode of transit for accessing Troy Middle School and Troy High School.

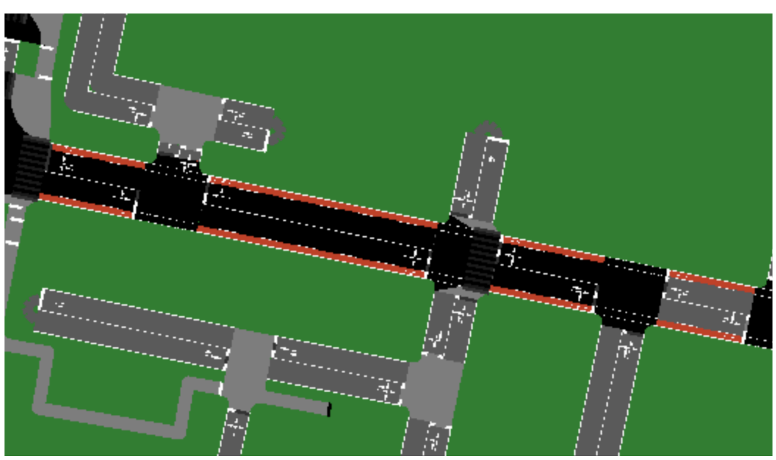

Figure 3. Proposed Red Bike Lanes on Burdett Ave.

To simulate the current and proposed conditions, the traffic simulation software Simulation of Urban Mobility (SUMO) and its supplemental platforms - Netedit and OSM Web Wizard - were used. Using OSM Web Wizard, the selected analysis zone - a rectangular region spanning past 10th St. to Burdett - was identified. Then, the Through Traffic Factor (TFF) and Count values for each vehicle type were determined. TFF was determined through a CDTC figure shown in Fig. (4) depicting peak morning and evening vehicle volumes.

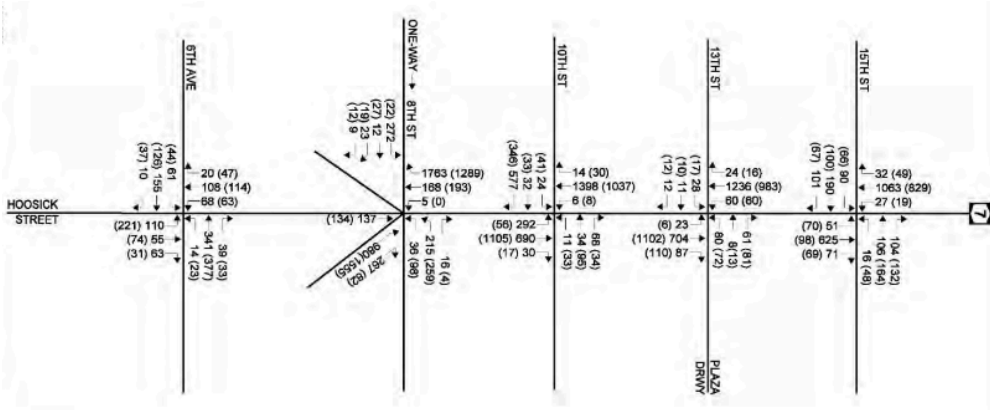

Figure 4. Vehicle Behavior at the Peak Morning and (Evening) Hour[2]

Using these figures, the TTF value was found by averaging Eq. (1) for 10th, 13th, and 15th St.:

$$\frac{East-West\,Traffic}{North-South\,Traffic} = TFF \tag{1}$$

To find the Count value Eq. (2) is used:

$$Count_{veh} = \frac{(Percent_{veh})\cdot(Total\,Volume)}{2\cdot(\#\,lanes)\cdot(Length\,of\,Hoosick\,St.)} \tag{2}$$

The vehicle makeup was determined using NYSDOT's Traffic Data Viewer, and the Total Volume was used from the Traffic Data Viewer as well. The length, set in km, of Hoosick St. was found using Google Earth data. After importing the Web Wizard data into SUMO, a simulation was run to determine data under current conditions. Then, the map was imported into Netedit to add bus lanes, road verges, and bike lanes. Afterward, the proposed street plan was simulated as well.

3. LIMITATIONS

3.1 Burdett Avenue Analysis

For Burdett Avenue, the scope of the change revolves around the interest of local residents, so it was difficult to find a quantitative improvement. Thus, the changes will be determined by the effect it has on Hoosick Street instead.

3.2 Behavior of Drivers

The lane removal will make an impact on the volume of vehicles passing through Troy, however these calculations are outside the scope of the project and cannot be determined. In reality, some traffic will be diverted to other routes like NY Route 2 or through Pleasantville north of Hoosick. In addition, it is difficult to determine the extent of which drivers in passenger vehicles will transition to alternate modes of transit. While the proposed changes will allow for changes in the bus system to improve significantly, this is reliant on a population that has been car dependent for the last few generations.

4. RESULTS

4.1 Initial Calculations

Shown in Table 1 are the vehicle counts used for the SUMO simulation. Bikes and Pedestrians were not included for the percentage as they do not use the roads.

Table 1. Count Values on Hoosick Street

Mode of Transportation	Hoosick Volume Percentage	Count
Passenger Car	80.14%	862
Trucks	18.78%	202
Bus	0.57%	6
Bicycle		2
Pedestrian		125

4.2 Current and Proposed Hoosick Street Results

Shown in Table 2 were the current and proposed Hoosick St. results. For vehicles and pedestrian statistics, speed, waiting time, and time loss was used:

Table 2. Current and Proposed Hoosick Street Statistics

Transportation Mode	Statistic	Current	Proposed
Car	Speed *(km/hr)*	3.13	3.61
	Duration *(s)*	823.4	399.23
	Waiting Time *(s)*	662.06	299.92
	Time Loss *(s)*	734.7	357.11
Pedestrian	Duration *(s)*	556.86	302.77
	Time Loss *(s)*	118.33	63.94

5. DISCUSSION AND CONCLUSION

As shown in Table 2, there is a distinct improvement after implementing the proposed changes. While there is a marginal increase in speed, duration, waiting time, and time loss for both cars and pedestrians decreased dramatically. This data assumed that the number of vehicles remained the same. If more citizens transition from car to buses both within Troy and from areas outside Troy, the proposed changes could be significantly more optimal than the current values. This would allow for even lower carbon emissions as the CDTA could allocate more electric buses for the 87 Route. These new changes should enhance the actions of drivers on Hoosick St due to the lower congestion, decreasing frustration which could lead to accidents on Hoosick. The installed greenverge incentivizes more people to walk on the sidewalks to locations on Hoosick, benefitting the Hillside and Frear Park residents.

Thus, the implementation of a dedicated bus lane and bike lanes on Burdett will have a significant effect on the welfare of the communities surrounding Hoosick Street. The changes implemented align with making Hoosick Street accessible by multiple modes of transportation, which will allow people throughout Troy to access commercial sites on or near Hoosick in a safe, affordable manner. The reduced time loss will decrease speeding and thus accidents on Hoosick Street. In the future, the increase of bus and pedestrian traffic on Hoosick will reduce carbon emissions, decreasing respiratory illnesses locally.

6. CITATIONS

[1]S. Simmons, “Elected officials and residents weigh in on the future of Hoosick Street and Road,” WAMC, Jan. 25, 2024. https://www.wamc.org/news/2024-01-25/elected-officials-and-residents-weigh-in-on-the-future-of-hoosick-street-and-road

[2]“Hoosick Hillside Study Final Report.” Available: https://www.capitalmpo.org/wp-content/CRTC/images/linkage_program/RenCoFinal/Hoosick_Hillside_Final_Report_sm.pdf

[3]“Route 87 - Hoosick St - RPI | www.cdta.org,” Cdta.org, 2024. https://www.cdta.org/schedules-route-detail?route_id=87 (accessed Apr. 20, 2025).

[4]“The Demographic Statistical Atlas of the United States - Statistical Atlas,” Statisticalatlas.com, 2025. https://statisticalatlas.com/block-group/New-York/Rensselaer-County/041300-5/Age-and-Sex (accessed Apr. 20, 2025).

[5] S. Rasca and N. Saeed, “Exploring the factors influencing the use of public transport by commuters living in networks of small cities and towns,” Travel Behaviour and Society, vol. 28, pp. 249–263, 2022, doi: https://doi.org/10.1016/j.tbs.2022.03.007.

7. APPENDIX

Data for Current and Proposed Hoosick Street Changes:

[1] msteenhuis, “GitHub - msteenhuis/Hoosick-Street: Current and Proposed changes to Hoosick Street,” GitHub, 2025. https://github.com/msteenhuis/Hoosick-Street

$$TFF = \frac{\frac{2596}{239}+\frac{2267}{215}+\frac{1192}{519}}{3} = 7.901 \tag{1.1}$$

$$Count_{Car} = \frac{(80.14\%)\cdot(2152\ Vehicles)}{2\cdot(2\ lanes)\cdot(0.8\ km)} = 862\ veh/hr \tag{2.1}$$